

NAME Nori Karthikey;

REGISTER NO 22BCI0052

DIGITAL ASSIGNMENT 2

FTK IMAGER EXERCISE QUESTIONS

Q1. What is FTK Imager primarily used for in digital forensics?

Ans:

FTK Imager is a key tool in digital forensics, used to create exact, forensically sound images of storage devices while maintaining evidence integrity for investigations and legal use. It allows investigators to capture data from computers, mobile devices, and other media without altering the source, preview and verify evidence before acquisition, and recover deleted or hidden data from unallocated space and file slack. The tool also generates hash values to confirm authenticity and supports both volatile (e.g., live memory) and non-volatile data acquisition from physical drives.

Q2. Which file formats can FTK Imager save forensic

images in? Ans:

FTK Imager supports saving forensic images in multiple popular formats, including Raw (dd/.001), EnCase Evidence File (E01), SMART, ISO/CUE, and Advanced Forensic Format (AFF).

Q3. How can you verify the integrity of an acquired image in

FTK Imager? Ans:

In FTK Imager, image integrity is verified using its built-in hash verification feature, which generates and compares cryptographic hashes for both the source device and the acquired image. By selecting the 'Verify images after they are created' option, the tool automatically computes hashes for each and includes them in a report. Matching values confirm the image is an exact replica of the original, while any mismatch indicates discrepancies that require further investigation.

Q4. What is the difference between Preview and Acquire Image options in FTK Imager?

Ans:

Preview –

The preview feature lets investigators examine the contents of a storage device without creating a full forensic image. It enables quick browsing of files to evaluate their potential evidentiary value before deciding on full acquisition. This process is non-destructive and leaves the original evidence unchanged.

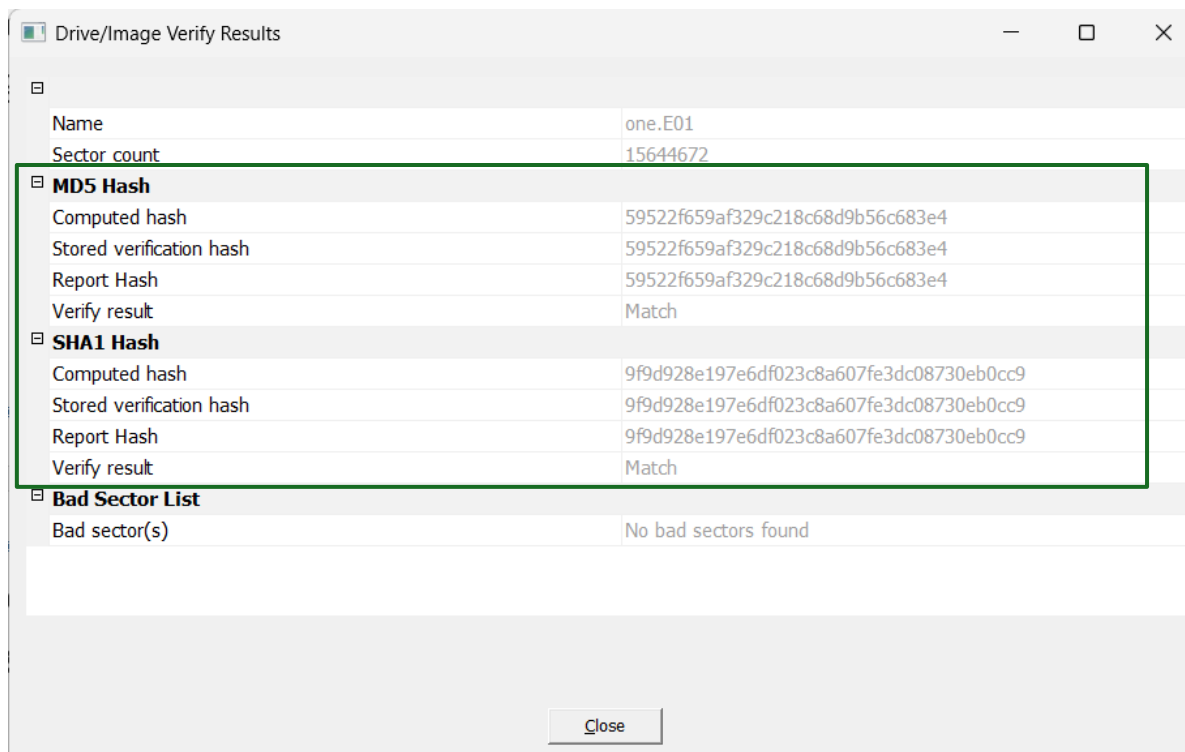
Acquire Image –

The acquire image option creates a forensically sound copy of the selected device or folder, capturing all data, including hidden, deleted, and system files. This image can later be mounted, previewed, or analyzed in forensic tools, allowing detailed examination without needing to access the original evidence again.

Q5. Where in FTK Imager can you find the hash values (MD5/SHA1/SHA256) of the evidence?

Ans:

Once the USB disk image is exported to the PC, its hash values



can be viewed. These can also be checked by right-clicking the evidence in the evidence tree.

Q6. What is the significance of the Evidence Item Information section when adding evidence?

Ans:

When adding evidence, the “Evidence Item Information”

section is used to record essential details about the item being added.

1. **Identification** – Captures the evidence name, type, and source to differentiate between multiple items in the same case, preventing confusion even if revisited years later.
2. **Chain of Custody** – Documents when, where, and by whom the evidence was obtained, forming part of the legal record to prove it was not altered, ensuring court admissibility.

3. Context and Analysis Support – Stores case numbers, investigator notes, and acquisition details, helping streamline the analysis by making the purpose and relevance of each item immediately clear.

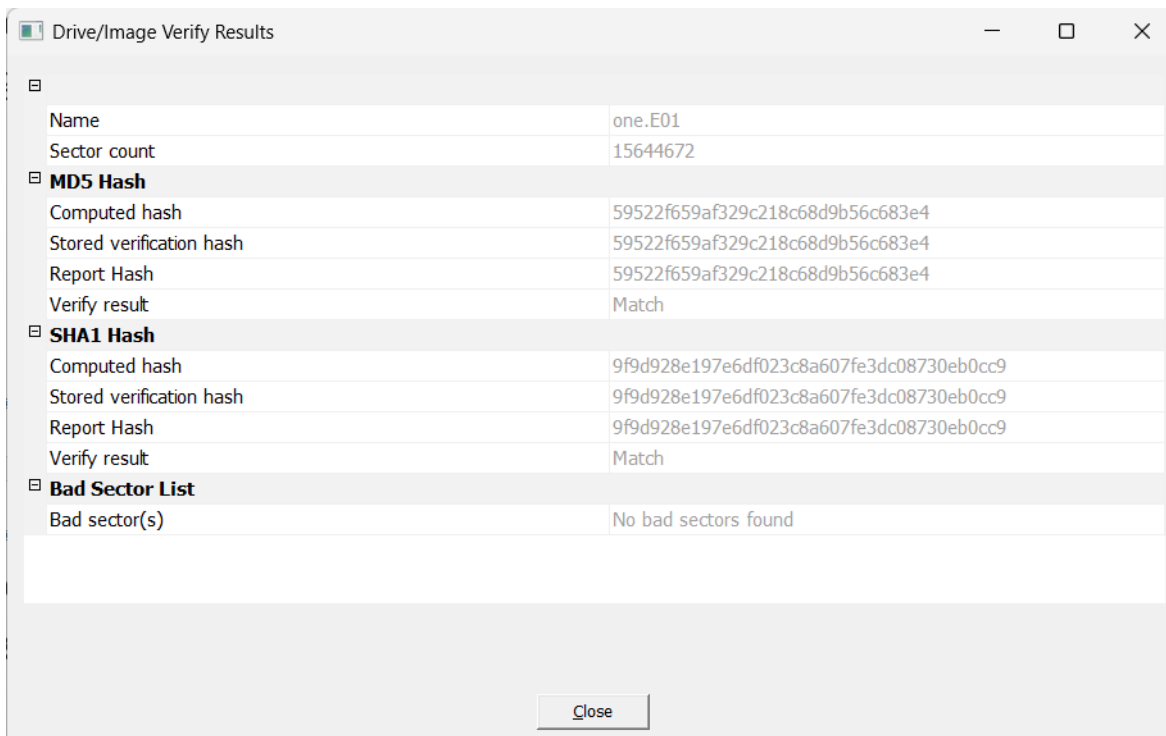
Q7. Use FTK Imager to acquire an image of a USB drive. Which image type did you select and why?

Ans:

In FTK Imager, the E01 format was chosen because it is a recognized standard in digital forensics and is admissible in court. It stores MD5/SHA1 hash values for integrity verification, supports lossless compression to save storage without altering evidence, and can embed metadata such as case details, examiner information, and notes. E01 also handles bad sectors effectively, ensuring maximum data recovery from damaged media, making it a reliable and forensically sound option for imaging a USB drive.

Q8. Generate both MD5 and SHA1 hashes for the acquired image. What are the results?

Ans:



MD5 Hash – 59522f659af329c218c68d9b56c683e4

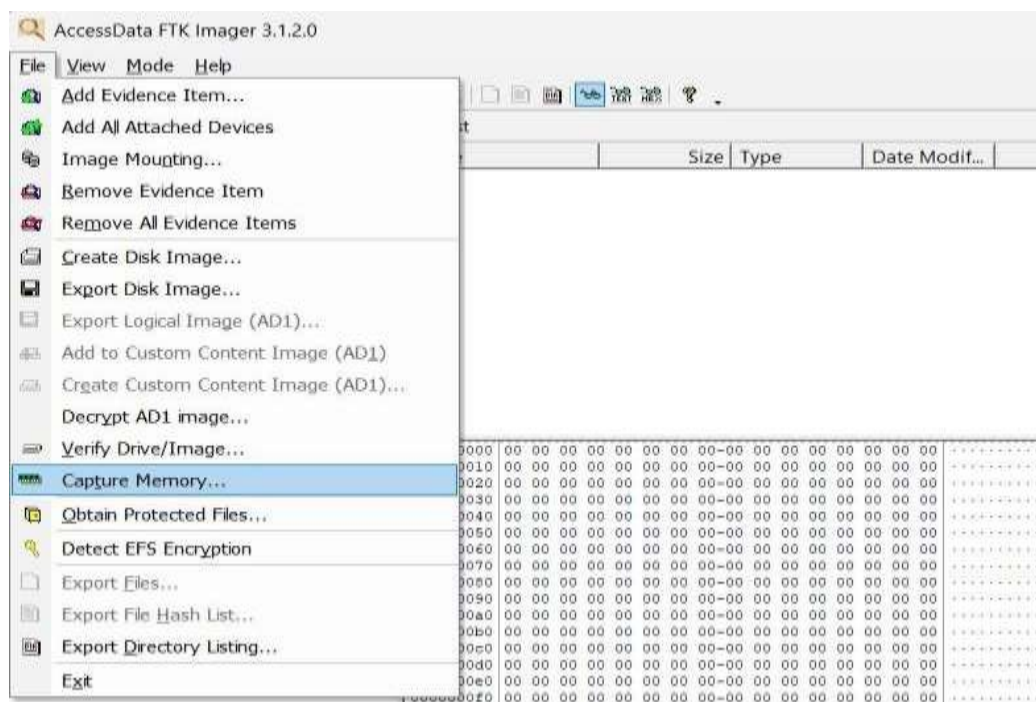
SHA1 Hash – 9f9d928e197e6df023c8a607fe3dc08730eb0cc9

The matching values of the computed, stored, and report hashes confirm that the image is an exact, unaltered copy of the original USB device.

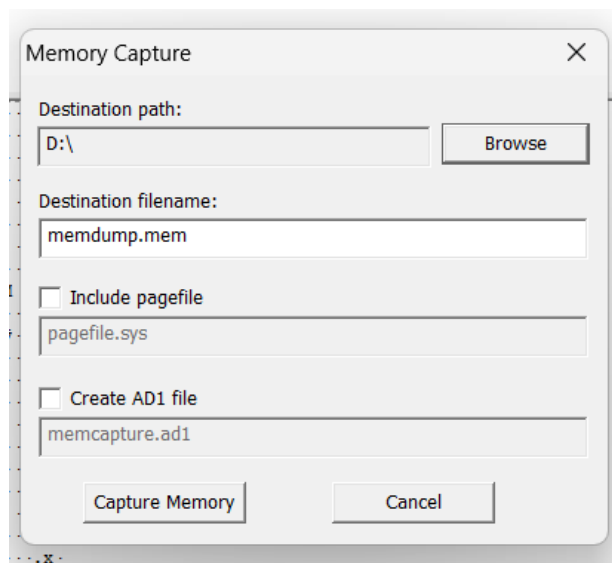
Q9. Capture a memory dump from a running system using FTK Imager. Which menu option did you use?

Ans:

File > Capture Memory



Click on capture memory after choosing path



A screenshot of a Windows-style dialog box titled "Memory Capture". The dialog has a close button (X) in the top right corner. It contains the following fields and controls:

- Destination path:** A text box containing "D:\", followed by a "Browse" button.
- Destination filename:** A text box containing "memdump.mem".
- Include pagefile:** An unchecked checkbox, followed by a text box containing "pagefile.sys".
- Create AD1 file:** An unchecked checkbox, followed by a text box containing "memcapture.ad1".
- Buttons:** "Capture Memory" and "Cancel" buttons at the bottom.

Q10. Add an existing E01 evidence file into FTKImager and locate the following:

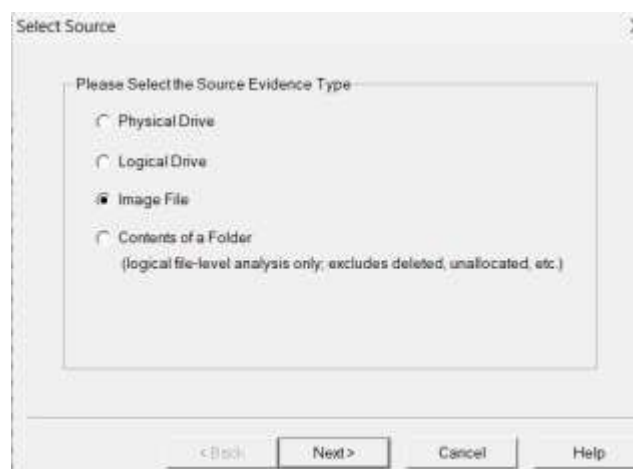
a. A Deleted file

b. A Hidden file

c. A file with a suspicious extension

Ans:

adding image file mantooth.E01



a. deleted file

	Name	Size	Type	Date Modif...
	Afrit	1	Directory	13/02/2008...
	Book of Nod	1	Directory	13/02/2008...
	Vampire Terminology.doc	77	Regular File	13/02/2008...
	Vampire.doc	72	Regular File	13/02/2008...

b. hidden file

Name	Size	Type	Date Modif...
desktop.ini	1	Regular File	06/03/2007...

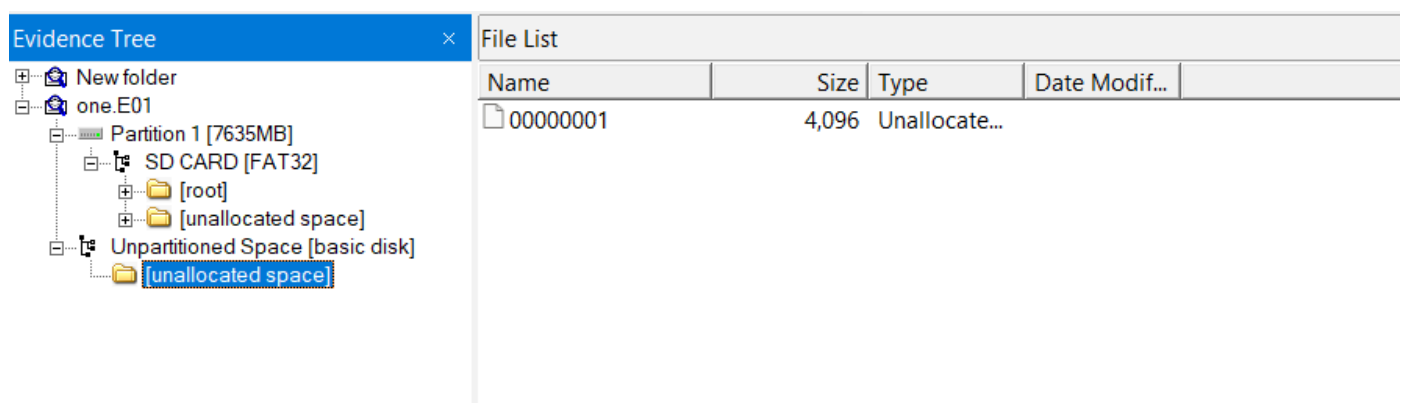
Name	Size	Type	Date Modif...
\$I30	4	NTFS Index ...	07/07/2007...
\$T...	1	Regular File	07/07/2007...

c. file with suspicious extension

Q11. In FTKImager, how can you view slack space and unallocated space?

Ans:

Slack space can be analyzed by opening a file in the hex viewer and identifying the unused bytes at the end of its allocated cluster. Unallocated space can be accessed by navigating to One.E01 > Unpartitioned Space > [unallocated space].



The file named '00000001', with a size of 4096 bytes, represents unallocated space.

Q12. What methods does FTKImager use to display deleted files, and how can you identify them?

Ans:

- **Identify Deleted Files:** Look for a red “X” icon beside the file name in the File List view.



- **Examine Unallocated Clusters:** In the Evidence Tree, navigate to *Unallocated Clusters* and inspect them in

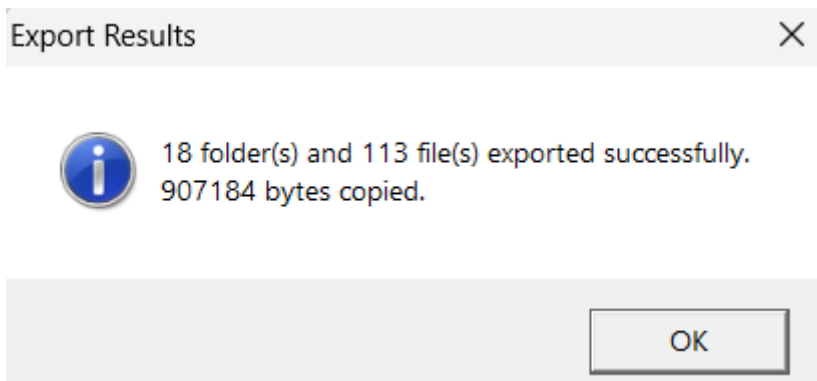
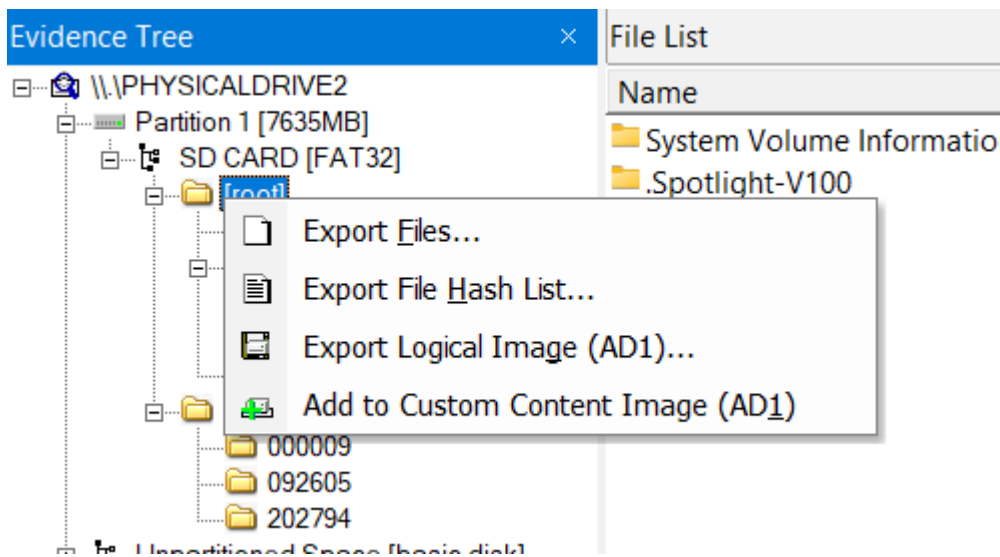
hex view for any file remnants.

- **Use the Preview Pane:** In some cases, portions or even the entire content of a deleted file may still be visible if its clusters have not been overwritten.

Q13. How would you extract a specific folder structure from an image file without altering evidence?

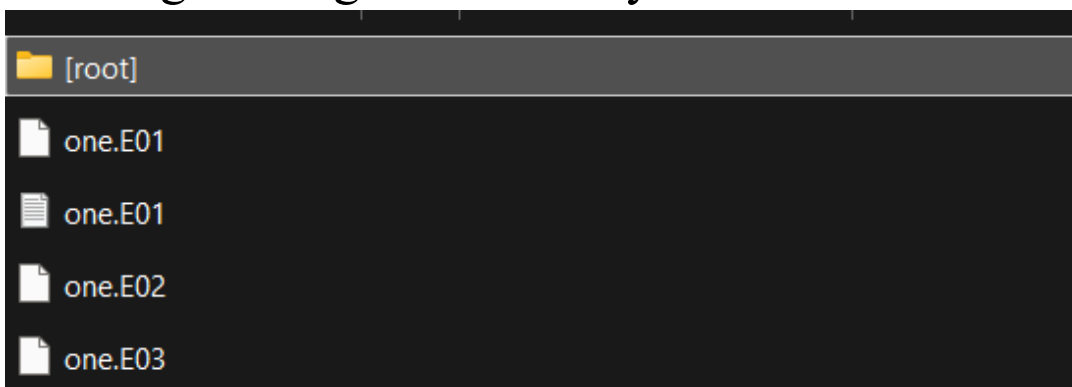
Ans:

Right click a folder > Export Files



Save to any directory

Viewing file in given directory



Ans:

A screenshot of the Spotlight-V100 software interface. The 'Tools' menu is open, and the 'Export File Hash List...' option is highlighted. The menu also includes 'Export Files...', 'Export Logical Image (AD1)...', and 'Add to Custom Content Image (AD1)'. The background shows a file tree with folders like 'Spotlight-V100' and '000009'.

Report is generated.

